Homework — 2.2.1 Programming Techniques

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 — set after the 2.2.1 lesson. Show traces/working. Code uses OCR Exam Reference Language.

Part A — This week's topic (~14 marks)

1. Trace the call `power(2, 4)` for the recursive function below. Complete the trace table, filling in the value **returned** at every level of the recursion. **[3]** *(AO2)*

```
function power(base, exp)
    if exp == 0 then
        return 1
    else
        return base * power(base, exp - 1)
    endif
endfunction
```

Call	Value returned
power(2, 4)	
power(2, 3)	
power(2, 2)	
power(2, 1)	
power(2, 0)	

2. The program below must add the numbers 1 to 10 using **count-controlled** iteration, then use **condition-controlled** iteration to keep asking until a number greater than 0 is entered. Complete the three missing keywords, labelled (a) to (c), using the word bank. Three items are not needed. **[3]** *(AO2)*

```
total = 0
(a) ________ i = 1 to 10
    total = total + i
(b) ________ i

do
    num = input("Enter a number greater than 0")
(c) ________ num > 0

print(total)
```

Word bank: for · next · until · while · endwhile · then

3. The program below is **intended** to output `0` and then `20`, but it contains **two** errors and actually outputs different values. (In OCR Exam Reference Language, a variable assigned inside a subroutine is **local** unless it is declared `global`.)

```
count = 8
total = 10

procedure reset()
    count = 0
endprocedure

procedure double(n:byVal)
    n = n * 2
endprocedure

reset()
double(total)
print(count)
print(total)
```

(a) State the two values the program **actually** prints, in order, justifying each with reference to **scope** or **parameter passing**. **[2]** *(AO2)*

(b) Rewrite the **two** lines that must change so that the program prints `0` and then `20`. **[2]** *(AO2)*

4. Write the letter of the correct definition in the Match column for each object-oriented term. Two definitions are not needed. **[2]** *(AO1)*

Term	Match
Class	
Object	
Encapsulation	
Inheritance	

Letter	Definition
A	A template that defines the attributes and methods shared by a type of object
B	A single instance of a class, created at run time with its own attribute values
C	Bundling attributes and methods together and keeping the attributes private, so they can only be accessed through the class's public methods
D	A class taking on the attributes and methods of a parent class, adding or overriding its own
E	A variable that can only be accessed inside the subroutine where it is created
F	The same method call producing different behaviour depending on the object it is called on

5. A program validates input by asking again for a value until the value entered is greater than 0. Explain why a `do ... until` loop is more suitable here than a `while` loop. **[2]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(2.1.1)* State what is meant by **abstraction** and give **one** reason it is useful when solving a problem. **[2]**

7. *(1.4.2)* A **stack** and a **queue** are both data structures. State the order in which items are removed from each (use the correct four-letter acronym). **[2]**

8. *(2.1.3)* Explain what is meant by **decomposition**. **[2]**